Graphs of transformed hyperbolas

1 Consider the function $y = -\frac{1}{4x}$.

a Complete the following table of values:

x	-3	-2	-1	1	2	3
y						

b Sketch the graph.

c In which quadrants does the graph lie?

2 Consider the function $y = \frac{2}{x+4}$.

a State the equation of the vertical asymptote.

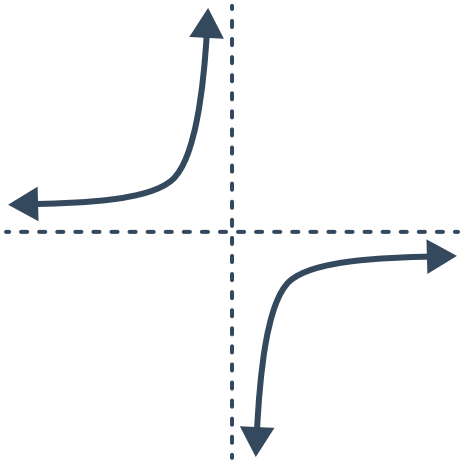
b State the equation of the horizontal asymptote.

c Sketch the graph of the function.

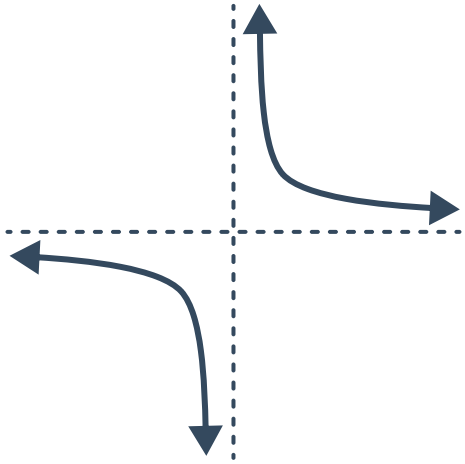
3 Consider the hyperbolic function $y = \frac{3}{x} - 3$.

a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii



b Which curve approaches positive and negative infinity more quickly: $y = \frac{1}{x}$ or $y = \frac{3}{x} - 3$?

c What are the equations of the vertical and horizontal asymptotes of $y = \frac{3}{x} - 3$?

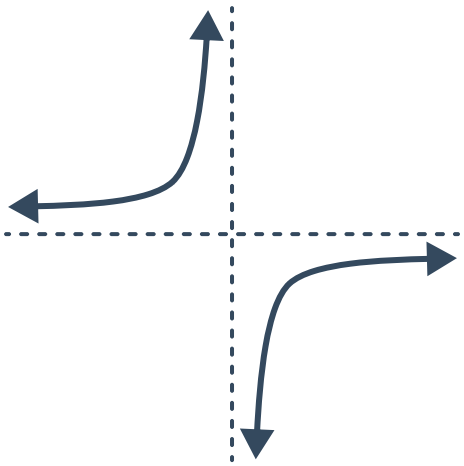
d Sketch the graph of $y = \frac{3}{x} - 3$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

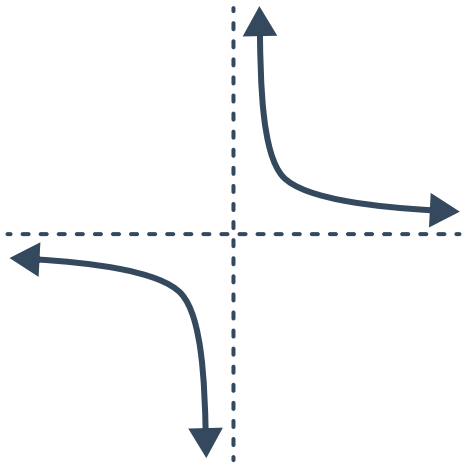
4 Consider the hyperbolic function $y = -\frac{3}{x} + 2$.

a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii



b Which curve approaches positive and negative infinity more quickly: $y = \frac{-1}{x}$ or $y = \frac{-3}{x} + 2$

c What are the equations of the vertical and horizontal asymptotes of $y = \frac{-3}{x} + 2$?

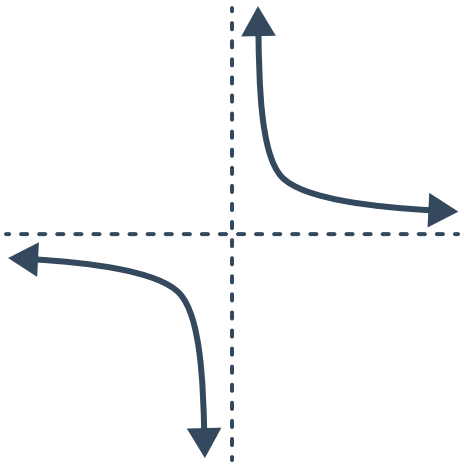
d Sketch the graph of $y = \frac{-3}{x} + 2$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

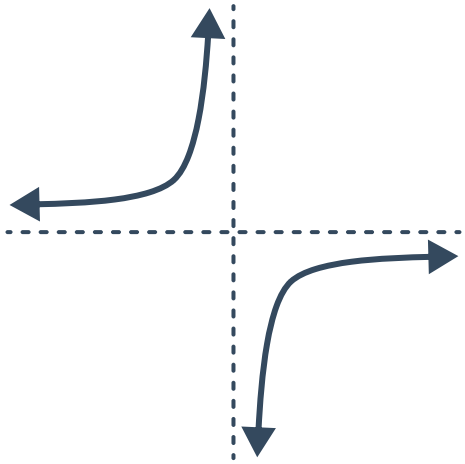
5 Consider the hyperbolic function $y = \frac{1}{5x} - 3$.

a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii



b Which curve approaches positive and negative infinity more quickly: $y = \frac{1}{x}$ or $y = \frac{1}{5x} - 3$

c What are the equations of the vertical and horizontal asymptotes of $y = \frac{1}{5x} - 3$?

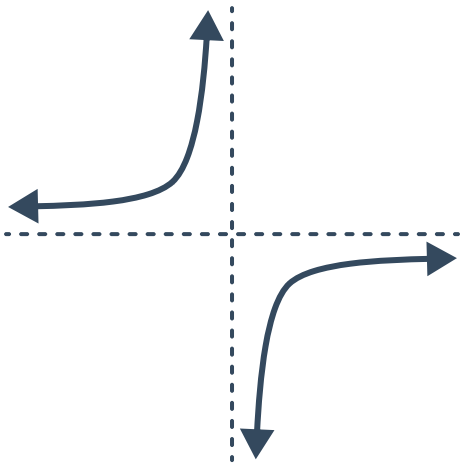
d Sketch the graph of $y = \frac{1}{5x} - 3$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

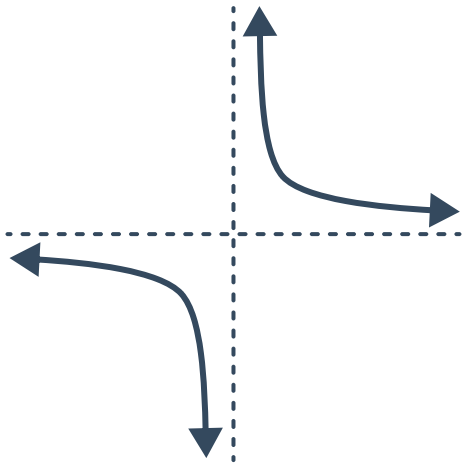
6 Consider the hyperbolic function $y = -\frac{1}{2x} + 2$.

a Determine whether the following graphs indicate the position of the hyperbola's branches relative to its asymptotes:

i



ii



b Which of these curves approach positive and negative infinity more quickly: $y = \frac{-1}{x}$ or $y = -\frac{1}{2x} + 2$

c What are the equations of the vertical and horizontal asymptotes of $y = -\frac{1}{2x} + 2$?

d Sketch the graph of $y = -\frac{1}{2x} + 2$.

7 Consider the function $y = \frac{1}{x-3}$.

a Complete the table of values:

x	1	2	$\frac{5}{2}$	$\frac{7}{2}$	4	5
y						

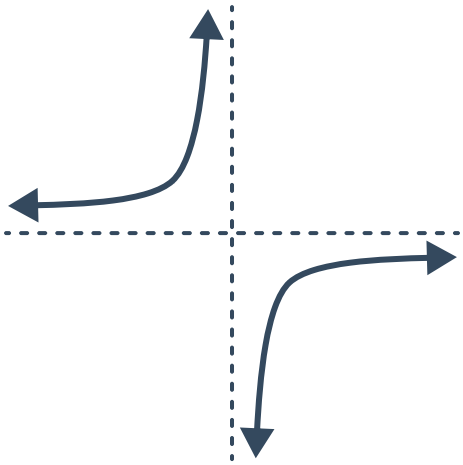
b Sketch the graph.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

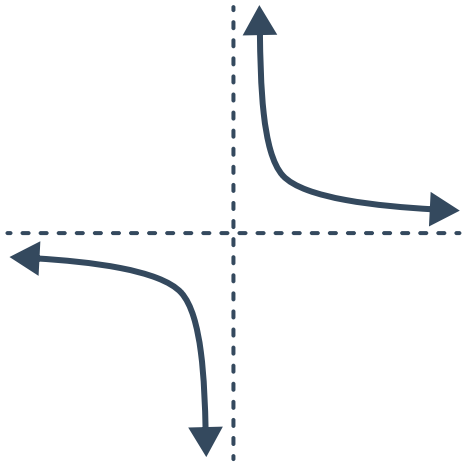
8 For each of the following hyperbolic functions:

i Determine which of the following graphs indicate the position of the hyperbola's branches relative to its asymptotes: **A** or **B**.

A



B



ii Write down the equations of the vertical and horizontal asymptotes.

iii Sketch the graph.

a $y = \dfrac{1}{4(x-2)}$

b $y = \dfrac{-2}{x-1}$

c $y = -\dfrac{1}{2(x-2)} + 2$

9 Consider the function $y = \dfrac{x-2}{x-4}$.

a Solve the following equation for a:

$$\frac{x-2}{x-4} = \frac{x-4+a}{x-4}$$

b Hence, express $y = \dfrac{x-2}{x-4}$ in the form $y = \dfrac{m}{x-h} + k$, for some values k and h.

c State the equation of the vertical asymptote.

d As x approaches ∞, what does y approach?

e Hence, state the equation of the horizontal asymptote.

f Sketch the graph of the function.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Consider the function $y = \dfrac{x-4}{x-3}$.

a Solve the following equation for a:

$$\frac{x-4}{x-3} = \frac{x-3-a}{x-3}$$

b Hence, express $y = \dfrac{x-4}{x-3}$ in the form $y = \dfrac{k}{x-3} + h$, for some values k and h.

c State the equation of the vertical asymptote.

d As x approaches ∞, what does y approach?

e Hence, state the equation of the horizontal asymptote.

f Sketch the graph of the function

11 Consider the function $y = \dfrac{4-x}{x-3}$.

a Express $y = \dfrac{4-x}{x-3}$ in the form $y = \dfrac{k}{x-3} + h$, for some values k and h.

b State the equation of the vertical asymptote.

c As x approaches ∞, what does y approach?

d Hence, state the equation of the horizontal asymptote.

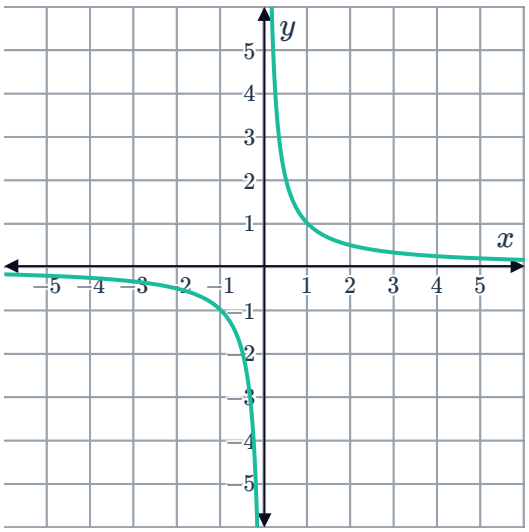
e Sketch the graph of the function.

Transformations and domain and range

12 Consider the graph of the hyperbola $y = \dfrac{1}{x}$:

a What would be the new equation if the graph was shifted upwards by 4 units?

b What would be the new equation if the graph was shifted to the right by 7 units?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13 Consider the following hyperbolas:

$$y = \frac{6}{x} \text{ and } y = \frac{6}{x} - 3$$

a What is the y value of $y = \frac{6}{x}$ corresponding to $x = -2$?

b What is the y value of $y = \frac{6}{x} - 3$ corresponding to $x = -2$?

c How is $y = \frac{6}{x}$ transformed to make $y = \frac{6}{x} - 3$?

14 Consider the following hyperbolas:

$$y = \frac{-1}{x} \text{ and } y = \frac{-1}{x-4}$$

a What value cannot be substituted for x in $y = \frac{-1}{x}$?

b In which quadrants does $y = \frac{-1}{x}$ lie?

c What value cannot be substituted for x in $y = \frac{-1}{x-4}$?

d In which quadrants does $y = \frac{-1}{x-4}$ lie?

e How can the graph of $y = \frac{-1}{x}$ be transformed to create the graph of $y = \frac{-1}{x-4}$?

15 A hyperbola has a domain of $x \in \mathbb{R}, x \neq 2$ and a range of $y \in \mathbb{R}, y \neq -3$. Determine whether the following could be the equation of the hyperbola:

a $y = \frac{1}{x-2} - 3$

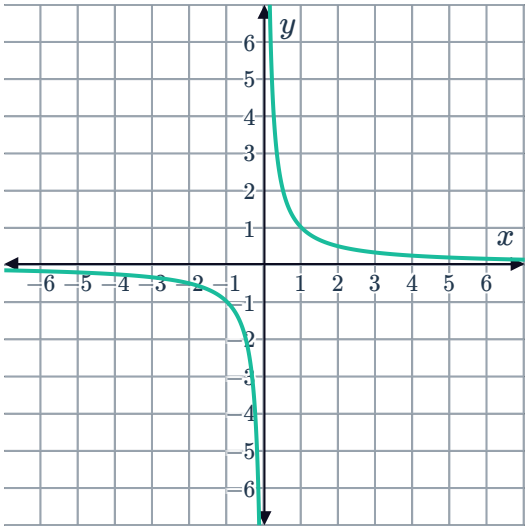
b $y = \frac{3}{x-2} + 3$

c $y = \frac{1}{3(x-2)} - 3$

d $y = \frac{1}{x-2} + 3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16 Consider the graph of $y = \frac{1}{x}$:



a How do we shift the graph of $y = \frac{1}{x}$ to get the graph of $y = \frac{1}{x} + 3$?

b How do we shift the graph of $y = \frac{1}{x}$ to get the graph of $y = \frac{1}{x+2}$?

c Sketch the graph of $y = \frac{1}{x} + 3$.

d Sketch the graph of $y = \frac{1}{x+2}$.

17 Consider the function $f(x) = \frac{3}{x}$.

a How can the graph of $f(x)$ be obtained from the graph of $y = \frac{1}{x}$?

b Sketch the graph of $f(x)$.

c What is the domain of $f(x)$?

d What is the range of $f(x)$?

e Is the function $f(x)$ increasing or decreasing over its domain?

18 Consider the function $f(x) = \frac{1}{x+4}$.

a How can the graph of $f(x)$ be obtained from the graph of $y = \frac{1}{x}$?

b Sketch the graph of $f(x)$.

c What is the domain of $f(x)$?

d What is the range of $f(x)$?

e Is the function $f(x)$ increasing or decreasing over its domain?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 Consider the function $y = \dfrac{2}{x+1}$.

- **a** State the domain of the function.
- **b** State the equation of the vertical asymptote.
- **c** Rearrange $y = \dfrac{2}{x+1}$ to make x the subject.
- **d** Hence, state the range of the function.
- **e** Hence, state the equation of the horizontal asymptote.
- **f** Sketch the graph of the function.

20 Consider the function $y = \dfrac{3}{x} + 2$.

- **a** State the domain of the function.
- **b** State the equation of the vertical asymptote.
- **c** Rearrange the equation to express x in terms of y.
- **d** State the range of the function.
- **e** Hence, state the equation of the horizontal asymptote.
- **f** Sketch the graph of the function.

21 Consider the function $y = -\dfrac{3}{x-1}$.

- **a** State the domain of the function.
- **b** State the equation of the vertical asymptote.
- **c** Rearrange the equation to express x in terms of y.
- **d** Hence, state the range of the function.
- **e** State the equation of the horizontal asymptote.
- **f** Sketch the graph of the function.

22 Consider the function $y = -\dfrac{3}{5(x+2)}$.

- **a** State the value of a that completes the domain of the function:
  Domain: $x \in \mathbb{R}, x \neq a$
- **b** State the equation of the vertical asymptote.
- **c** As x approaches ∞, what value does y approach?
- **d** Hence, state the equation of the horizontal asymptote.
- **e** State the value of b that completes the range of the function:
  Range: $x \in \mathbb{R}; y \neq b$
- **f** Sketch the graph of the function.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Applications

23 The time, t, taken by a typist to type up a document is inversely proportional to his typing speed, s. That is, the quicker the typing speed, the less time it will take. If it takes a typist 20 minutes to type a particular document, typing at a speed of 61 words per minute:

- **a** Find the constant of variation k.
- **b** How long (in minutes) will it take a typist with a typing speed of 30.5 words per minute to type up the document?

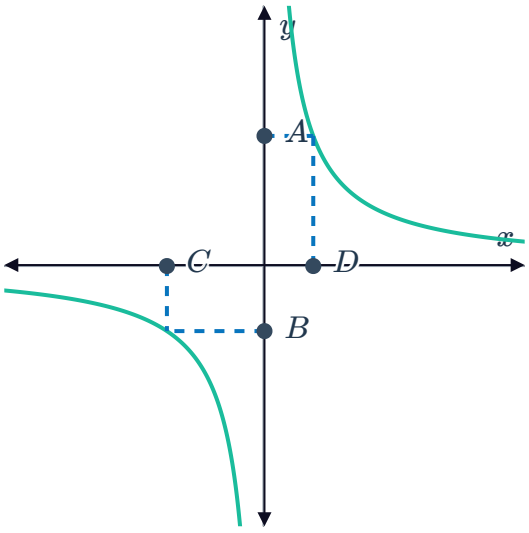
24 The rent, electricity, telephone bill and other expenses for a flat cost a total of $\$490$ per week. These expenses are shared equally between the tenants of the flat.

- **a** How much will each occupant pay if the flat is shared by two people?
- **b** Let the number of occupants be x, and the cost paid by each occupant be y. Write a formula that relates the two variables.
- **c** Use this equation to fill in the following table of values. Round your answers correct to two decimal places where necessary.

x	1	2	3	4	5	6
y						

- **d** Sketch curve for the equation.
- **e** From the graph, what type of relationship exists between x and y?

25 A graph of the hyperbola $y = \dfrac{10}{x}$ is shown: Given points $C\,(-4, 0)$ and $D\,(2, 0)$, find the length of interval AB.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647